

CS 362 : Artificial Intelligence

Course Code : CS 362

Course Name : Artificial Intelligence

L-T-P-C : 3-0-0-3

Intended for : 3rd and 4th year B.Tech. students

Prerequisite : CS 212 or an equivalent course

Mutual Exclusion:

Approval: 54th BoA

Course Contents

- **Introduction to AI:** (3 Hours) Overview, historical perspective, Turing test, agents.
- **Search Methods and Optimization:** Problem representation, State Space Search, A* Algorithm, AO* search, Minimax and alphabeta pruning, AI in games. (8 Hours)
- **Logic and Automated Reasoning:** Propositional Logic, Predicate Calculus, Resolution Refutation, Formal Systems, Soundness, Consistency, Completeness. (7 Hours)
- **Planning and Constraint Satisfaction:** Blocks World, STRIPS, Constraint Satisfaction Problems, Basics of Probabilistic Planning, Waltz Algorithm. (6 Hours)
- **Knowledge Representation and Reasoning:** Semantic Net, Frames, Scripts, Conceptual Dependency, Ontologies, Basics of Semantic Web, Intelligent Question Answering. (7 Hours)
- **Learning and Uncertainty:** Learning from Examples, Decision Trees, Neural Nets, Bayesian Theory, Fuzzy Logic, Nonmonotonic Logic, Default Reasoning, Case-based Reasoning. (6 Hours)
- **Advanced Topics in AI:** Introduction to Computer Vision, Expert Systems, Natural Language Processing, Robotics, Hidden Markov Models, Reinforcement Learning. (5 Hours)

Textbooks:

1. Russell, S., & Norvig, P., **Artificial Intelligence: A Modern Approach**, 4th Edition, Pearson Education, 2020.
2. Poole, D., & Mackworth, A., **Artificial Intelligence: Foundations of Computational Agents**, 2nd Edition, Cambridge University Press, 2017.

References:

1. Murphy, K. P., **Machine Learning: A Probabilistic Perspective**, MIT Press, 2012.
2. Luger, G. F., **Artificial Intelligence: Structures and Strategies for Complex Problem Solving**, 6th Edition, Pearson Education, 2008.